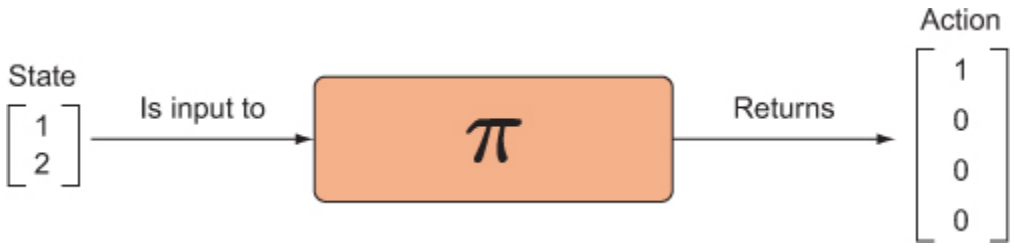
we may not end up taking the same action every time. In figure 4.3 we feed our function the state, which is (1,2), and the output is a vector of probabilities corresponding to each action. If this was a Gridworld agent, for example, the agent would have a 0.50 probability of going up, no chance of going down, a 0.25 probability of going left, and a 0.25 probability of going right.

If the environment is stationary, which is when the distribution of states and rewards is constant, and we use a deterministic strategy, we'd expect the probability distribution to eventually converge to a *degenerate probability distribution*, as shown in figure 4.4. A degenerate probability distribution is a distribution in which all the probability mass is assigned to a single potential outcome. When dealing with discrete probability distributions as we do in this book, all the probabilities must sum to 1, so a degenerate distribution is one where all outcomes are assigned 0 probability except for one, which is assigned 1.

Figure 4.4. A deterministic policy function, often represented by the Greek character pi, takes a state and returns a specific action to take, unlike a stochastic policy, which returns a probability distribution over actions.



Early in training we want the distribution to be fairly uniform so that we can maximize exploration, but over the course of training we want the distribution to converge on the optimal actions, given a state. If there is only one optimal action for a state, we'd expect to converge toward a degenerate distribution, but if there are two equally good actions, then we would expect the distribution to have two *modes*. A mode of a probability distribution is just another word for a "peak."

I forget ... What's a probability distribution?